**SZABIST**
University

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**ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT**

**Total Marks:_____**

**Obtained Marks: _____**

# Computing Vision

## Lab Task # 03

**Submitted To:** **Mr. Muhammad Azhar**

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**Part 1: Import Required Libraries**
**Part 2: Upload Image from Local System**
**Part 3: Read the Uploaded Image**
**Part 4: Check if the Image Loaded Successfully**
**Part 5: Convert BGR to RGB for Display**
**Part 6: Convert Image to Grayscale**
**Part 7: Resize the Image to 300x300 Pixels**
**Part 8: Display Original, Grayscale, and Resized Images**
**Part 9: Save Grayscale and Resized Images**
**Part 10: Confirmation Message**

```
[1]: import cv2
     import matplotlib.pyplot as plt
     import numpy as np

[2]: img = cv2.imread("D:/23108106/AA1.jpg")
     if img is None:
         print("Error: Image not found or unable to read.")

[3]: img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

[4]: gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

[5]: resized = cv2.resize(img, (300, 300))

[6]: resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)

[7]: plt.figure(figsize=(12, 4))
     plt.subplot(1, 3, 1)
     plt.title("Original Image")
     plt.imshow(img_rgb)
     plt.axis('off')
     plt.subplot(1, 3, 2)
     plt.title("Grayscale Image")
     plt.imshow(gray, cmap='gray')
     plt.axis('off')
     plt.subplot(1, 3, 3)
     plt.title("Resized Image")
     plt.imshow(resized_rgb)
     plt.axis('off')
     plt.tight_layout()
     plt.show()
```

```
[8]: cv2.imwrite('D:/23108106/gray_sample.jpg', gray)
     cv2.imwrite('D:/23108106/resized_sample.jpg', resized)
     print("Images saved as 'gray_sample.jpg' and 'resized_sample.jpg")

     Images saved as 'gray_sample.jpg' and 'resized_sample.jpg
```

**1. Image Quality Enhancer for Social Media**

**You are developing a basic photo editing tool for a mobile app that enhances images before posting to social media.**

**Task:**

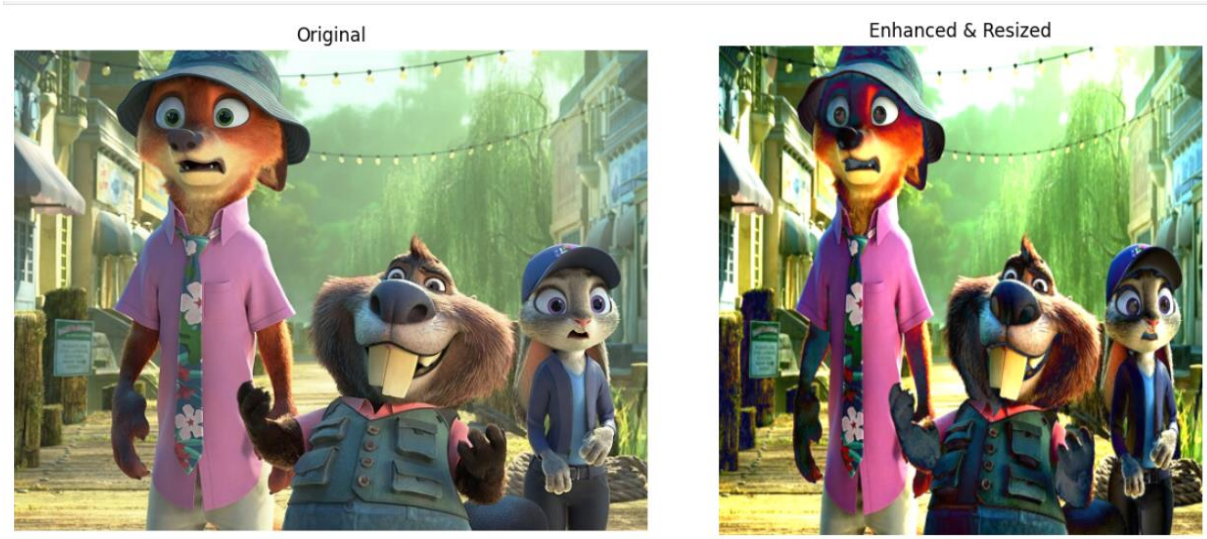
**Write a Python program using OpenCV to:**

- **Read an image.**
- **Automatically adjust brightness and contrast.**
- **Resize it to optimal dimensions (e.g., 1080x1080).**
- **Save the enhanced version.**

```
[9]: # 1. Image Quality Enhancer for Social Media
img2 = cv2.imread("D:/23108106/AA2.jpeg")
```

```
[10]: def auto_brightness_contrast(image, clip_hist_percent=25):
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    hist = cv2.calcHist([gray],[0],None,[256],[0,256]).flatten()
    acc = np.cumsum(hist)
    clip = clip_hist_percent * (acc[-1] / 100.0) / 2.0
    min_gray = np.searchsorted(acc, clip)
    max_gray = np.searchsorted(acc, acc[-1] - clip)
    alpha = 255 / (max_gray - min_gray + 1e-5)
    beta = -min_gray * alpha
    return cv2.convertScaleAbs(image, alpha=alpha, beta=beta)
enhanced = auto_brightness_contrast(img2)
resized = cv2.resize(enhanced, (1080, 1080))
```

```
[11]: fig, ax = plt.subplots(1, 2, figsize=(12, 5))
ax[0].imshow(cv2.cvtColor(img2, cv2.COLOR_BGR2RGB))
ax[0].set_title("Original")
ax[1].imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
ax[1].set_title("Enhanced & Resized")
for a in ax:
    a.axis('off')
plt.tight_layout()
plt.show()
```



**2. Security Camera Grayscale Archiver**

**Your client wants a system that converts all images from a CCTV feed to grayscale for efficient storage.**

**Task:**

**Simulate this by:**

- **Read a single color image from disk.**
- **Convert the image to grayscale.**
- **Save the grayscale image with a new filename**

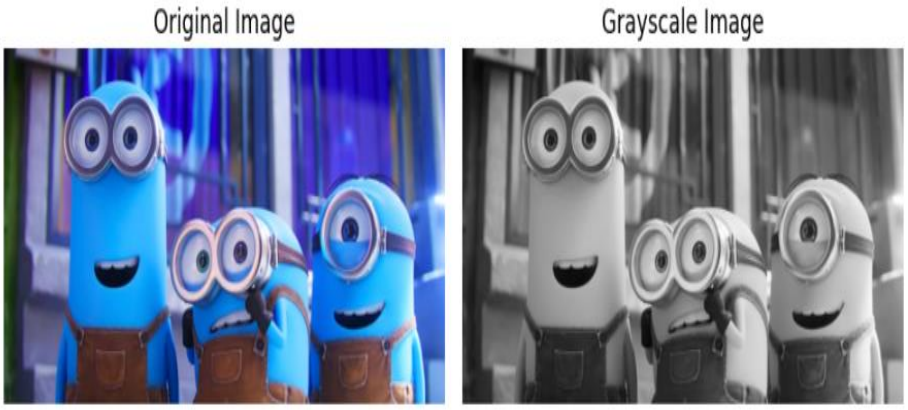
[12]:
```
# 2. Security Camera Grayscale Archiver
color_image = cv2.imread("D:/23108106/AA3.jpg")
if color_image is None:
    print("Error: Unable to read the image.")
```

[13]:
```
gray_image = cv2.cvtColor(color_image, cv2.COLOR_BGR2GRAY)
```

[14]:
```
cv2.imwrite('D:/23108106/gray_sample2.jpg', gray_image)
print("Image saved")
```

```
Image saved
```

[15]:
```
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.title("Original Image")
plt.imshow(color_image)
plt.axis('off')
plt.subplot(1, 3, 2)
plt.title("Grayscale Image")
plt.imshow(gray_image, cmap='gray')
plt.axis('off')
plt.tight_layout()
plt.show()
```



**3. Image Comparison for Duplicates**

**Build a tool that checks whether two uploaded images are duplicates or not.**

**Task:**

- **Read two images.**
- **Resize both to the same dimensions.**
- **Compare their pixel similarity using Mean Squared Error (MSE) or Structural Similarity Index (SSIM).**

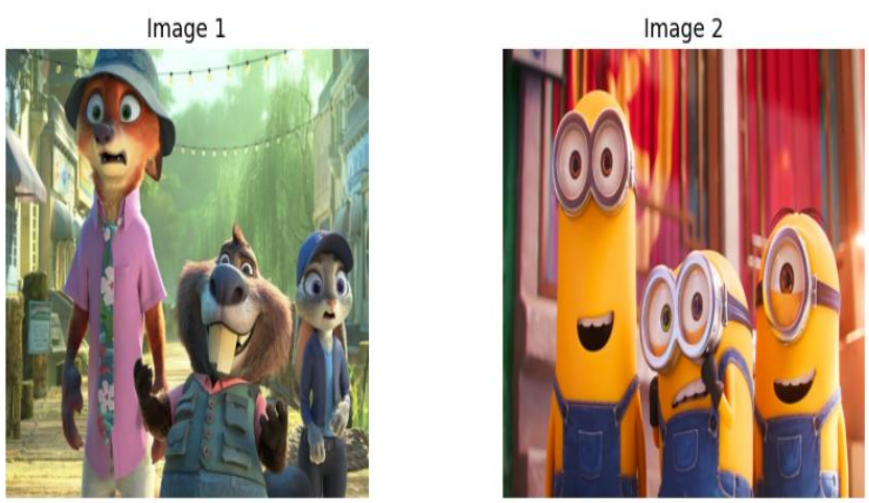
[16]:
```
# 3. Image Comparison for Duplicates
img2 = cv2.imread("D:/23108106/AA2.jpeg")
img3 = cv2.imread("D:/23108106/AA3.jpg")
img2 = cv2.resize(img2, (500, 500))
img3 = cv2.resize(img3, (500, 500))
```

[17]:
```
gray1 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(img3, cv2.COLOR_BGR2GRAY)
```

[18]:
```
from skimage.metrics import structural_similarity as ssim
mse = np.mean((gray1 - gray2) ** 2)
ssim_score, _ = ssim(gray1, gray2, full=True)
print(f"MSE: {mse:.2f}")
print(f"SSIM: {ssim_score:.4f}")
```

```
MSE: 106.20
SSIM: 0.1992
```

[19]:
```
fig, ax = plt.subplots(1, 2, figsize=(10, 4))
ax[0].imshow(cv2.cvtColor(img2, cv2.COLOR_BGR2RGB))
ax[0].set_title("Image 1")
ax[1].imshow(cv2.cvtColor(img3, cv2.COLOR_BGR2RGB))
ax[1].set_title("Image 2")
for a in ax:
    a.axis('off')
plt.show()
```



**4. Color Filter for Object Detection**

**Design a system to detect only red-colored objects in an image.**

**Task:**

- **Read an image.**
- **Convert to HSV color space.**
- **Use masking to isolate red objects.**

```
[20]: # 4. Color Filter for Object Detection
image = cv2.imread("D:/23108106/AA1.jpg")
if image is None:
    print("Error: Couldn't read the image.")
else:
    hsv_image = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
    lower_red1 = np.array([0, 120, 70])
    upper_red1 = np.array([10, 255, 255])
    lower_red2 = np.array([170, 120, 70])
    upper_red2 = np.array([180, 255, 255])
    mask1 = cv2.inRange(hsv_image, lower_red1, upper_red1)
    mask2 = cv2.inRange(hsv_image, lower_red2, upper_red2)
    red_mask = mask1 + mask2
    red_detected = cv2.bitwise_and(image, image, mask=red_mask)
    plt.figure(figsize=(12, 6))
    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis('off')
    plt.subplot(1, 2, 2)
    plt.imshow(cv2.cvtColor(red_detected, cv2.COLOR_BGR2RGB))
    plt.title("Red Objects Detected")
    plt.axis('off')
    plt.tight_layout()
    plt.show()
```

**5. Blurring License Plates for Privacy**
**You are tasked with protecting personal data in dashcam footage.**
**Task:**

- **Detect and blur rectangular regions that resemble license plates (manual ROI for now).**
- **Save and display the protected version.**

```
[21]: # 5. Blurring License Plates for Privacy
      image = cv2.imread("D:/23108106/AA4.jpg")

[22]: license_plate_rois = [(240, 250, 170, 95)]

[23]: height, width = image.shape[:2]

      for (x, y, w, h) in license_plate_rois:
          if x < 0 or y < 0 or x + w > width or y + h > height:
              print(f"Skipping ROI {(x, y, w, h)} because it's outside the image boundaries.")
              continue
          roi = image[y:y+h, x:x+w]
          blurred_roi = cv2.GaussianBlur(roi, (51, 51), 0)
          image[y:y+h, x:x+w] = blurred_roi

[24]: plt.figure(figsize=(10, 6))
      plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
      plt.title("Protected Dashcam Image (License Plates Blurred)")
      plt.axis('off')
      plt.show()
```

Protected Dashcam Image (License Plates Blurred)

```
]: cv2.imwrite("D:/23108106/protected_dashcam.jpg", image)
   print("image saved")

   image saved
```

**Case Studies:**

**1. Custom Image Watermarking Tool**

**A photographer wants to automatically add a watermark to all their work.**

**Task:**

- **Read the original image and a watermark (transparent PNG).**
- **Overlay the watermark at the bottom-right corner with adjustable transparency.**

[26]:
```
# Custom Image Watermarking Tool
img = cv2.imread("D:/23108106/AA1.jpg")
watermark = cv2.imread("D:/23108106/AA6.png",cv2.IMREAD_UNCHANGED)
if img is None or watermark is None:
    print("Error: Unable to read images.")
```

[27]:
```
h_img, w_img = img.shape[:2]
scale_width = int(w_img * 0.3)
scale_height = int(watermark.shape[0] * (scale_width / watermark.shape[1]))
watermark = cv2.resize(watermark, (scale_width, scale_height))
h_wm, w_wm = watermark.shape[:2]
x = (w_img - w_wm) // 2
y = (h_img - h_wm) // 2
overlay = watermark[:, :, :3]
alpha = watermark[:, :, 3] / 255.0
opacity = 0.8
for c in range(3):
    img[y:y+h_wm, x:x+w_wm, c] = (
        (1 - alpha * opacity) * img[y:y+h_wm, x:x+w_wm, c] +
        (alpha * opacity) * overlay[:, :, c]
    )
```

[28]:
```
plt.figure(figsize=(8,6))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Watermarked Image")
plt.axis("off")
plt.show()
```

Watermarked Image

[29]:
```
cv2.imwrite("D:/23108106/watermarked_output.jpg", img)
print("Watermarked image saved.")
```

```
Watermarked image saved.
```

## 2. Basic OCR Pipeline (Conceptual)

**Design the preprocessing part of an OCR (Optical Character Recognition) tool.**

**Task:**

- **Load a scanned document image.**
- **Convert to grayscale, remove noise, and apply thresholding to prepare for OCR.**

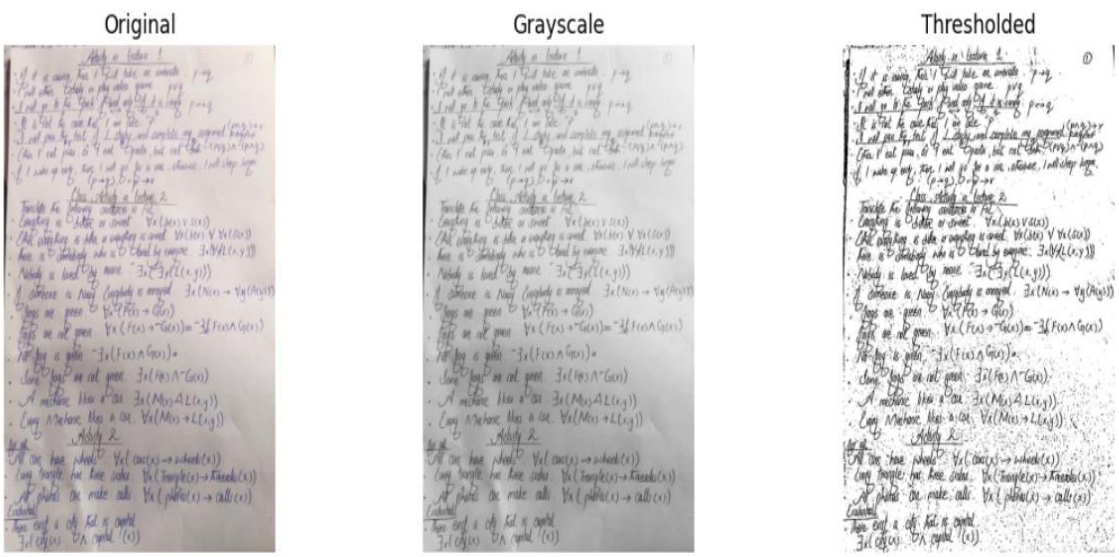
[30]:
```
# Basic OCR Pipeline (Conceptual)
img = cv2.imread("D:/23108106/AA5.jpeg")
if img is None:
    print("Error: Image not found.")
```

[31]:
```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

[32]:
```
denoised = cv2.medianBlur(gray, 3)
```

[33]:
```
thresh = cv2.adaptiveThreshold(denoised, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY, 11, 2)
```

[34]:
```
plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.title("Original")
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis("off")

plt.subplot(1,3,2)
plt.title("Grayscale")
plt.imshow(gray, cmap='gray')
plt.axis("off")

plt.subplot(1,3,3)
plt.title("Thresholded")
plt.imshow(thresh, cmap='gray')
plt.axis("off")
plt.tight_layout()
plt.show()
```




[35]:
```
cv2.imwrite("D:/23108106/ocr_ready.jpg", thresh)
print("OCR-ready image saved.")
```

```
OCR-ready image saved.
```

**3. Collage Generator Using Python and OpenCV**
**Create a 2x2 photo collage generator.**
**Task:**

- **Allow users to upload 4 images.**
- **Resize and arrange them into a single image grid (2 rows x 2 columns).**
- **Save the final collage.**

[36]:
```
# Collage Generator Using Python and OpenCV
img1 = cv2.imread("D:/23108106/AA1.jpg")
img2 = cv2.imread("D:/23108106/AA2.jpeg")
img3 = cv2.imread("D:/23108106/AA3.jpg")
img4 = cv2.imread("D:/23108106/AA4.jpg")
```

[37]:
```
img1 = cv2.resize(img1, (400,400))
img2 = cv2.resize(img2, (400,400))
img3 = cv2.resize(img3, (400,400))
img4 = cv2.resize(img4, (400,400))
```

[38]:
```
top_row = np.hstack((img1, img2))
bottom_row = np.hstack((img3, img4))
collage = np.vstack((top_row, bottom_row))
```

[39]:
```
plt.figure(figsize=(8,8))
plt.title("2x2 Collage")
plt.imshow(cv2.cvtColor(collage, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.show()
```

2x2 Collage

[40]:
```
cv2.imwrite("D:/23108106/collage_output.jpg", collage)
print("Collage saved successfully.")
```

```
Collage saved successfully.
```

**4. Color-to-Grayscale Converter for Print Shops**
**A print shop needs a tool to convert all client images to grayscale for black-and-white printing.**
**Task:**

- **Read and process a batch of images from a directory.**
- **Convert each to grayscale.**
- **Save all with a _bw suffix.**

---

[41]:
```
# Color-to-Grayscale Converter for Print Shops
import os

input_folder = "D:/23108106"
output_folder = "D:/23108106"

if not os.path.exists(output_folder):
    os.makedirs(output_folder)

for file in os.listdir(input_folder):
    if file.endswith((".jpg", ".png", ".jpeg")):
        img = cv2.imread(os.path.join(input_folder, file))
        if img is None:
            continue
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        name, ext = os.path.splitext(file)
        new_name = name + "_bw" + ext
        cv2.imwrite(os.path.join(output_folder, new_name), gray)
        print(f"{new_name} saved.")

print("All images converted to grayscale.")
```

```
AA1_bw.jpg saved.
AA2_bw.jpeg saved.
AA3_bw.jpg saved.
AA4_bw.jpg saved.
AA5_bw.jpeg saved.
AA6_bw.png saved.
collage_output_bw.jpg saved.
gray_sample_bw.jpg saved.
gray_sample2_bw.jpg saved.
ocr_ready_bw.jpg saved.
protected_dashcam_bw.jpg saved.
resized_sample_bw.jpg saved.
watermarked_output_bw.jpg saved.
All images converted to grayscale.
```

## 5. Noise Removal in Scanned Images

**You are working on improving quality of scanned documents with smudges and noise.**

**Task:**

- **Read a noisy image.**
- **Apply Gaussian blur or median filtering.**
- **Use thresholding to improve readability.**

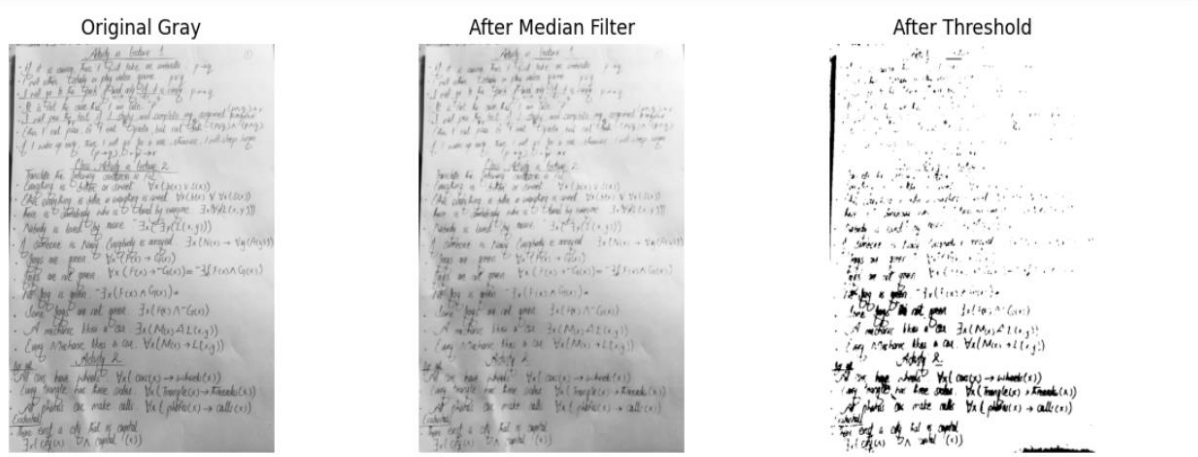
[42]:
```
# Noise Removal in Scanned Images
img = cv2.imread("D:/23108106/AA5.jpeg")
if img is None:
    print("Error: Image not found.")
```

[43]:
```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

[44]:
```
median = cv2.medianBlur(gray, 5)
```

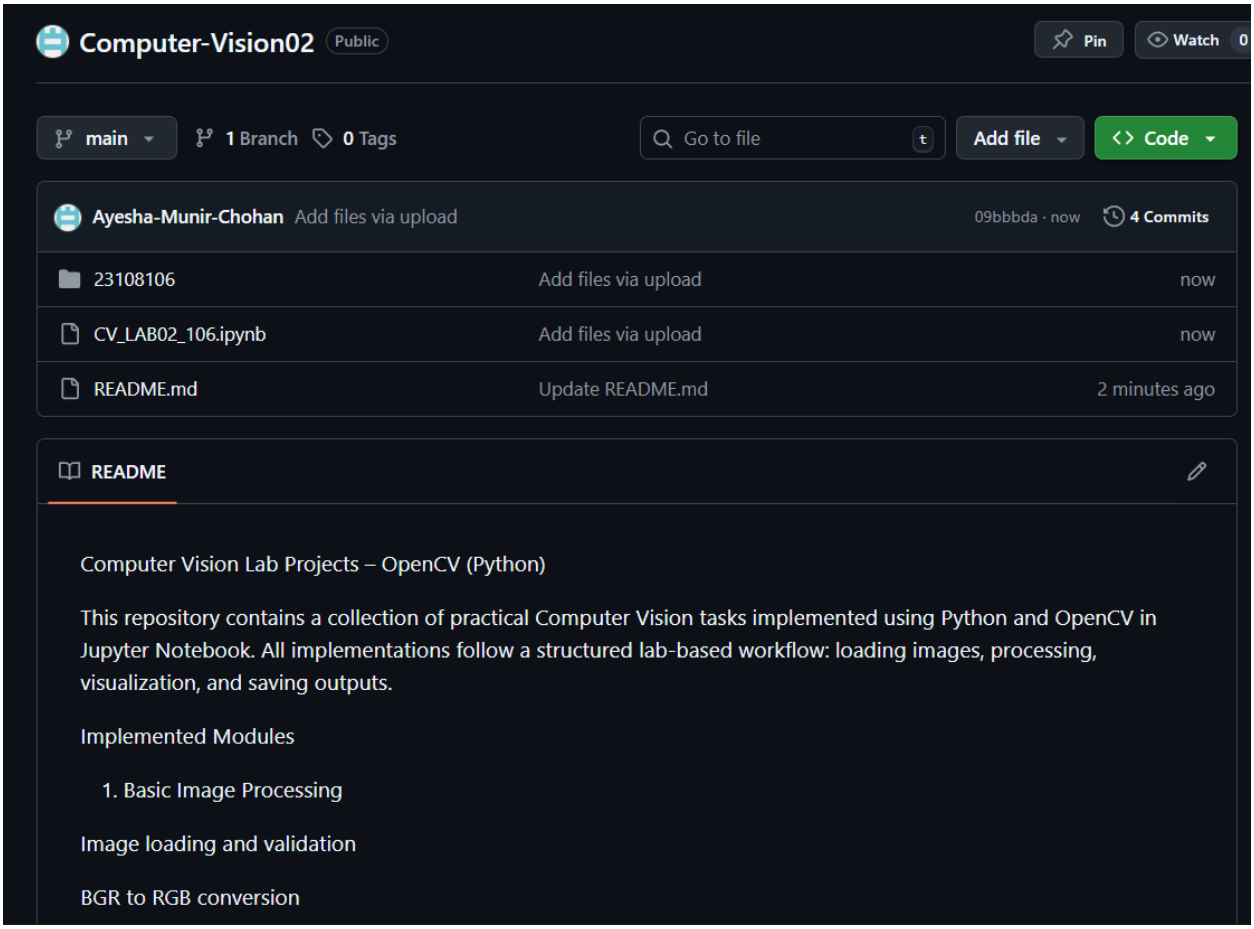
[45]:
```
_, thresh = cv2.threshold(median, 150, 255, cv2.THRESH_BINARY)
```

[46]:
```
plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.title("Original Gray")
plt.imshow(gray, cmap='gray')
plt.axis("off")

plt.subplot(1,3,2)
plt.title("After Median Filter")
plt.imshow(median, cmap='gray')
plt.axis("off")

plt.subplot(1,3,3)
plt.title("After Threshold")
plt.imshow(thresh, cmap='gray')
plt.axis("off")
plt.tight_layout()
plt.show()
```




47]:
```
cv2.imwrite("D:/23108106/cleaned_output.jpg", thresh)
print("Cleaned image saved.")
```

```
Cleaned image saved.
```




**https://github.com/Ayesha-Munir-Chohan/**